Convex Optimization: CSE203B Project Report

Ethan Chen - A69043007
Gayathri Rajesh - A69040613
Tejal Kulkarni - A69043100
Tomasz Frelek - A69042552

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1 Headline

i. **Title:** Convex Optimization for Electric Vehicle Charging Station Placement under Transportation and Power Grid Constraints.

ii. **Team Members:** Ethan Chen, Gayathri Rajesh, Tejal Kulkarni, Tomasz Frelek

iii. **Task Assignment:**

- All team members will conduct literature review, discuss modeling choices, and collaborate on writing the final report.tex.
- Ethan worked on the Convex Relaxation approach and consolidation of results.
- Gayathri worked on the Mixed Integer Linear Programming approach and consolidation of results.
- Tejal worked on generating the synthetic dataset, the greedy heuristic approach and the corresponding results.
- Tomasz worked on formulating the primal and dual problems, KKT conditions and analysis of results.

2 Introduction

2.1 Motivation

The global transition towards sustainable transportation has accelerated the adoption of Electric Vehicles (EVs) as a primary solution to reduce greenhouse gas emissions and fossil fuel dependency. However, the widespread integration of EVs is heavily constrained by the availability and accessibility of charging infrastructure. Inadequate or poorly planned charging networks lead to "range anxiety" among drivers—the fear that a vehicle has insufficient range to reach its destination—which remains a significant barrier to mainstream EV adoption [3].

Conversely, the uncoordinated installation of high-capacity charging stations poses a severe threat to existing urban power grids. Simultaneous high-demand charging, especially during peak hours, can lead to local transformer overloads, severe voltage deviations, and broader grid instability [1]. Therefore, the Electric Vehicle Charging Station Placement Problem (EVCSPP) is a complex intersection of transportation logistics and power systems engineering.

An effective charging station network must balance a critical trade-off: maximizing user convenience by minimizing the travel distance from demand nodes to stations, while strictly respecting the physical capacity limits of the local electrical grid. Motivated by this dual requirement, this project seeks to formulate an optimization framework that simultaneously addresses traffic demand fulfillment and electrical grid constraints. Developing a computationally efficient model to solve this joint placement problem is essential for urban planners to deploy infrastructure that is both driver-friendly and grid-resilient.

2.2 Previous Works

The literature surrounding EVCSPP has evolved significantly over the past decade. Early foundational studies focused primarily on the geographic distribution of stations and traffic flows. Jia et al. [6] provided early methodologies for the optimal siting and sizing of EV charging stations to meet spatial demand. Building on the operational mechanics of placement, Lam et al. [1] developed a rigorous and widely recognized mathematical formulation of the EVCSPP, formally establishing its computational complexity (NP-hardness) and proposing initial solution frameworks.

As the problem definitions matured, researchers recognized the necessity of integrating traffic behavior with electrical infrastructure. He et al. [3] made a foundational contribution by formulating the relationship between traffic flow routing and station deployment as a convex optimization problem, demonstrating how strategic placement can alleviate range anxiety. Concurrently, Wang et al. [4] introduced a vital joint modeling approach that explicitly coupled local power grid operational limits with dynamic traffic demand, highlighting the risks of grid bottlenecks.

A persistent challenge in these joint models, however, is the inherent non-convexity of power grid physics and the combinatorial nature of discrete placement decisions. A critical breakthrough in power systems optimization was provided by Low [7], who demonstrated how non-convex Alternating Current (AC) power flow equations could be mathematically relaxed into convex Second-Order Cone Programs (SOCP). Leveraging these advanced relaxation techniques, Zhang et al. [5] successfully formulated a convexified model to solve the joint planning of EV charging stations and power distribution networks, circumventing the combinatorial explosion faced by traditional mixed-integer solvers.

Our work builds directly upon these advancements by synthesizing the rigorous problem formulation of Lam et al. [1] with the joint traffic-grid considerations of Wang et al. [4], ultimately employing the convex relaxation frameworks championed by Low [7] and Zhang et al. [5] to achieve a highly scalable and computationally tractable placement model.

2.3 Intended Contributions

Our primary contribution is the development of a computationally tractable, globally solvable convex optimization framework for the joint EVCSPP.

Our specific methodological contributions are threefold:

1. **Integer and Topological Relaxation:** We relax the strict binary placement variables ($x_j \in \{0, 1\}$) into continuous capacity investments ($x_j \geq 0$) and eliminate non-convex quadratic network connectivity requirements, enabling gradient-based optimization.

2. **Grid-Aware Bounding:** Drawing on the convexification of AC power flows [7], we introduce strictly linear bounds to ensure that the aggregate installed charging capacity does not exceed the maximum allowable power limits ($P_{\max,k}$) of local grid zones.

3. **Algorithmic Scalability:** By mapping the synthesized problem into a strictly convex domain, we bypass the exponential time complexity of traditional branch-and-bound MILP solvers. This allows our formulation to be solved efficiently in polynomial time $O((MN)^{2.5})$ using standard interior-point methods, yielding provably near-optimal infrastructure plans.

2.4 Organization of the Paper

The remainder of this report is organized as follows. Section III formally defines the mathematical Statement of the Problem, detailing the primal formulation, its dual, and the associated Karush-Kuhn-Tucker (KKT) conditions. Section IV describes the synthetic dataset generation and outlines the three solution approaches: a Greedy Heuristic, a Mixed-Integer Linear Program (MILP), and our proposed Convex Relaxation. Section V presents the computational results and comparative analysis. Finally, Section VI concludes the paper and discusses avenues for future work.

3 Statement of the Problem

3.1 Primal Formulation

Before defining our proposed model, it is essential to contextualize the classical Electric Vehicle Charging Station Placement Problem (EVCSPP). Traditionally, this problem is formulated as a Mixed-Integer Non-Linear Program (MINLP) [1].

The core decision relies on discrete facility location. Let $x_j \in \{0, 1\}$ be a binary decision variable indicating whether a charging station is constructed at candidate location j, and let $y_{ij} \geq 0$ be the continuous fraction of EV charging demand from node i routed to station j. The classical objective minimizes the combined fixed installation costs and travel distance utility:

$$\min_{x,y} \quad \sum_{j=1}^{N} f_j x_j + \lambda \sum_{i=1}^{M} \sum_{j=1}^{N} d_{ij} y_{ij}$$

While this objective function is linear, attempting to solve the classical MINLP exactly results in an exponential combinatorial explosion for large-scale urban networks. To achieve a computationally tractable and scalable model, we introduce a convex relaxation that systematically addresses the three severe non-convexities present in the original formulation:

- **Binary Capacity Limits (Integer Relaxation):** The classical model enforces strict integer capacity limits ($\sum D_i y_{ij} \leq C_j x_j$ where $x_j \in \{0, 1\}$). We relax this discrete requirement by transforming the binary choice into a continuous, non-negative variable ($x_j \geq 0$). Instead of a hard "build or do not build" decision, x_j now represents a continuous capacity investment level, enabling gradient-based optimization.

- **Topological Network Requirements (Quadratic Relaxation):** The standard EVCSPP requires built stations to form a connected operational subgraph, historically enforced through non-convex quadratic terms (e.g., $x_i x_k$) to model network flows. Our formulation relaxes this strict topological requirement entirely. We preserve linearity by allowing the continuous fraction variable (y_{ij}) to dictate demand routing without requiring a rigid, interconnected physical station graph.

- **AC Power Flow (Linear Grid Bounds):** Real-world power grids are governed by Kirchhoff's circuit laws and non-convex sinusoidal power flow equations. We apply a standard convex relaxation to these physics [7]. In our simplified formulation, this is abstracted linearly as $\sum_{j \in \text{Zone } k} x_j \leq P_{\max,k}$, ensuring the aggregate capacity in a local grid zone strictly respects transformer limits without requiring non-linear AC circuit calculations.

Mathematical Model

Based on the problem requirements, the objective is to minimize a weighted sum of the installation cost and the travel distance utility. We define the following variables and parameters:

- x_j: The capacity or continuous investment level at station j, where a station represents a candidate location where EV charging infrastructure can be installed.

- y_{ij}: The fraction of demand from node i served by station j, where node i represents a demand location such as a residential area, commercial zone, or traffic hub generating EV charging demand.

- f_j: Fixed installation cost at station j.

- d_{ij}: Distance or travel time between node i and station j.

- λ: Weighting factor balancing cost and utility.

- D_i: Total demand at node i.

- C_j: Maximum capacity scaling factor for station j.

- $P_{\max,k}$: Maximum power limit for local grid zone k.

The convex relaxed primal problem is formulated as:

$$
\begin{aligned}
\min_{x,y} \quad & \sum_{j=1}^{N} f_j x_j + \lambda \sum_{i=1}^{M} \sum_{j=1}^{N} d_{ij} y_{ij} \\
\text{subject to} \quad & \sum_{j=1}^{N} y_{ij} = 1, \quad \forall i \in \{1, \ldots, M\} \quad \text{(Demand Fulfillment)} \\
& \sum_{i=1}^{M} D_i y_{ij} - C_j x_j \leq 0, \quad \forall j \in \{1, \ldots, N\} \quad \text{(Capacity Constraint)} \\
& \sum_{j \in \text{Zone } k} x_j - P_{\max,k} \leq 0, \quad \forall k \in \{1, \ldots, K\} \quad \text{(Grid Constraint)} \\
& x_j \geq 0, \quad y_{ij} \geq 0, \quad \forall i, j \quad \text{(Non-negativity)}
\end{aligned}
$$

3.2 Dual Formulation

To derive the dual problem, we first construct the Lagrangian by introducing dual variables for each constraint. Let $\nu_i \in \mathbb{R}$ be the dual variables for the equality constraints. Let $\mu_j \geq 0$ and $\gamma_k \geq 0$ be the dual variables for the capacity and grid inequality constraints, respectively. Let $k(j)$ denote the grid zone k to which station j belongs.

The Lagrangian $L(x, y, \nu, \mu, \gamma)$ is:

$$
\begin{aligned}
L(x, y, \nu, \mu, \gamma) = & \sum_{j=1}^{N} f_j x_j + \lambda \sum_{i=1}^{M} \sum_{j=1}^{N} d_{ij} y_{ij} \\
& + \sum_{i=1}^{M} \nu_i \left(1 - \sum_{j=1}^{N} y_{ij} \right) \\
& + \sum_{j=1}^{N} \mu_j \left(\sum_{i=1}^{M} D_i y_{ij} - C_j x_j \right) \\
& + \sum_{k=1}^{K} \gamma_k \left(\sum_{j \in \text{Zone } k} x_j - P_{\max,k} \right)
\end{aligned}
$$

Next, we rearrange the Lagrangian to group the terms associated with the primal variables x_j and y_{ij}:

$$
\begin{aligned}
L(x, y, \nu, \mu, \gamma) = & \sum_{i=1}^{M} \nu_i - \sum_{k=1}^{K} \gamma_k P_{\max,k} \\
& + \sum_{j=1}^{N} x_j \left(f_j - C_j \mu_j + \gamma_{k(j)} \right) \\
& + \sum_{i=1}^{M} \sum_{j=1}^{N} y_{ij} \left(\lambda d_{ij} - \nu_i + D_i \mu_j \right)
\end{aligned}
$$

The dual function $g(\nu, \mu, \gamma) = \inf_{x \geq 0, y \geq 0} L(x, y, \nu, \mu, \gamma)$. For this infimum to be bounded from below, the coefficients of the non-negative primal variables must be non-negative. This yields the following dual constraints:

$$
\begin{aligned}
f_j - C_j \mu_j + \gamma_{k(j)} \geq 0, \quad & \forall j \in \{1, \ldots, N\} \\
\lambda d_{ij} - \nu_i + D_i \mu_j \geq 0, \quad & \forall i, j
\end{aligned}
$$

When these conditions are met, the infimum occurs at $x = 0$ and $y = 0$, leaving only the constant terms. Thus,

the dual problem is to maximize this bounded dual function:

$$
\begin{aligned}
\max_{\nu,\mu,\gamma} \quad & \sum_{i=1}^{M} \nu_i - \sum_{k=1}^{K} P_{\max,k}\gamma_k \\
\text{subject to} \quad & \nu_i - D_i\mu_j \leq \lambda d_{ij}, \quad \forall i \in \{1,\dots,M\}, \forall j \in \{1,\dots,N\} \\
& C_j\mu_j - \gamma_{k(j)} \leq f_j, \quad \forall j \in \{1,\dots,N\} \\
& \mu_j \geq 0, \quad \forall j \in \{1,\dots,N\} \\
& \gamma_k \geq 0, \quad \forall k \in \{1,\dots,K\} \\
& \nu_i \in \mathbb{R}, \quad \forall i \in \{1,\dots,M\}
\end{aligned}
$$

3.3 Karush-Kuhn-Tucker Conditions

The KKT conditions provide the necessary and sufficient conditions for optimality in our convex relaxed model. Let x^*, y^* be the optimal primal variables and ν^*, μ^*, γ^* be the optimal dual variables. To handle the non-negativity constraints explicitly, we introduce dual multipliers $\alpha_j \geq 0$ for $-x_j \leq 0$ and $\beta_{ij} \geq 0$ for $-y_{ij} \leq 0$.

The four sets of KKT conditions are derived as follows:

1. Stationarity

The gradient of the Lagrangian with respect to the primal variables must be zero at the optimum. Taking the partial derivatives of $L(x, y, \nu, \mu, \gamma)$ gives:

$$
\begin{aligned}
\nabla_{x_j} L &= f_j - C_j\mu_j^* + \gamma_{k(j)}^* - \alpha_j^* = 0, \quad \forall j \in \{1,\dots,N\} \\
\nabla_{y_{ij}} L &= \lambda d_{ij} - \nu_i^* + D_i\mu_j^* - \beta_{ij}^* = 0, \quad \forall i \in \{1,\dots,M\}, \forall j \in \{1,\dots,N\}
\end{aligned}
$$

2. Primal Feasibility

The optimal primal variables must satisfy the original constraints of the problem:

$$
\begin{aligned}
\sum_{j=1}^{N} y_{ij}^* &= 1, \quad \forall i \in \{1,\dots,M\} \\
\sum_{i=1}^{M} D_i y_{ij}^* - C_j x_j^* &\leq 0, \quad \forall j \in \{1,\dots,N\} \\
\sum_{j \in \text{Zone } k} x_j^* - P_{\max,k} &\leq 0, \quad \forall k \in \{1,\dots,K\} \\
x_j^* \geq 0, \quad y_{ij}^* &\geq 0, \quad \forall i, j
\end{aligned}
$$

3. Dual Feasibility

The optimal dual variables corresponding to the inequality constraints must be non-negative:

$$
\begin{aligned}
\mu_j^* &\geq 0, \quad \forall j \in \{1,\dots,N\} \\
\gamma_k^* &\geq 0, \quad \forall k \in \{1,\dots,K\} \\
\alpha_j^* &\geq 0, \quad \forall j \in \{1,\dots,N\} \\
\beta_{ij}^* &\geq 0, \quad \forall i \in \{1,\dots,M\}, \forall j \in \{1,\dots,N\}
\end{aligned}
$$

4. Complementary Slackness

The product of each dual variable and its corresponding inequality constraint must evaluate to zero. This implies that a dual variable can only be strictly positive if its corresponding constraint is strictly active (tight):

$$\mu_j^* \left(\sum_{i=1}^{M} D_i y_{ij}^* - C_j x_j^* \right) = 0, \quad \forall j$$

$$\gamma_k^* \left(\sum_{j \in \text{Zone } k} x_j^* - P_{\max,k} \right) = 0, \quad \forall k$$

$$\alpha_j^* x_j^* = 0 \implies x_j^* \left(f_j - C_j \mu_j^* + \gamma_{k(j)}^* \right) = 0, \quad \forall j$$

$$\beta_{ij}^* y_{ij}^* = 0 \implies y_{ij}^* \left(\lambda d_{ij} - \nu_i^* + D_i \mu_j^* \right) = 0, \quad \forall i, j$$

Interestingly, the complementary slackness conditions provide valuable economic insights. For instance, if a charging station j operates below its maximum capacity ($\sum D_i y_{ij}^* < C_j x_j^*$), its corresponding shadow price μ_j^* must be 0. Similarly, if a grid zone k has excess power capacity, $\gamma_k^* = 0$, indicates that power limits are not driving the marginal cost in that specific zone.

4 Approaches

4.1 Sample Dataset

Since publicly available datasets containing detailed information on electric vehicle charging demand, candidate station locations, and grid constraints are limited, synthetic datasets are generated to evaluate the proposed optimization approaches. The generation process is designed to approximate realistic urban demand patterns and infrastructure limitations.

The study area is modeled as a two–dimensional spatial region representing an urban environment. Within this region, demand nodes and candidate charging stations are generated. Demand nodes represent locations where EV charging demand originates, such as residential areas or commercial districts, while candidate stations represent potential sites where charging infrastructure can be installed.

To simulate realistic demand patterns, demand nodes are generated using a clustered spatial model. Cluster centers are randomly sampled within the region, and demand nodes are drawn from Gaussian distributions around these centers. This creates localized regions of high demand density, reflecting the non-uniform distribution of EV charging demand observed in real cities. Each demand node i is assigned a demand value D_i sampled from a predefined range.

Candidate charging stations are distributed uniformly across the region. Each station j is associated with an installation cost f_j and a capacity C_j. Installation costs are slightly higher for stations located near the center of the region to reflect higher infrastructure costs in central urban areas.

Distances between demand nodes and candidate stations are computed using Euclidean distance to form the matrix d_{ij} used in the optimization model. To incorporate electrical infrastructure constraints, the region is divided into grid zones, and each station is assigned to a zone based on its location. For each zone k, a maximum allowable capacity $P_{\max,k}$ is generated as a fraction of the total station capacity within that zone.

To evaluate scalability, datasets of different sizes (small, medium, and large) are generated. Beyond simply increasing the total count of nodes and stations, the underlying parameter ranges also scale up with the dataset size. Specifically, larger instances exhibit higher demand per node, alongside proportionally higher installation costs and larger individual station capacities. Furthermore, the maximum allowable grid capacity ($P_{\max,k}$) becomes significantly more relaxed in larger models. For each size, we generated 5 datasets via random seed, and evaluated our algorithms on the same 15 total datasets.

The spatial distribution of demand nodes and candidate charging stations generated using this procedure is illustrated in Figure 1.

4.2 Evaluated Problem Formulations

To fully contextualize the approaches evaluated in this study, it is necessary to understand how the mathematical formulation of the EVCSPP dictates the choice of computational solvers. The problem transitions through three distinct phases of mathematical complexity:

Table 1: Parameter scaling ranges across synthetic dataset sizes.

Parameter	Small	Medium	Large
Demand per node	8-20	8-30	10-35
Install cost	73-161	70-205	82-234
Station capacity	21-42	26-69	30-83
Grid $P_{\max}$	53-78	129-238	123-900

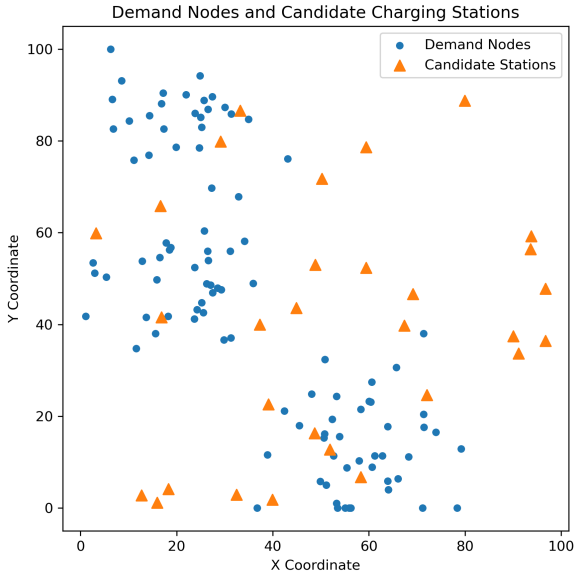

Figure 1: Spatial distribution of demand nodes and candidate charging stations in the synthetic dataset.

1. The Original Problem: MINLP

As established in Section 3, the pure real-world problem is a Mixed-Integer Non-Linear Program (MINLP). It combines discrete binary decisions (building stations or not) with highly non-linear physical constraints, such as quadratic topological routing and sinusoidal AC power flow physics. Because MINLPs are formally NP-hard and contain non-convex continuous relaxations, specialized spatial branch-and-bound solvers experience severe combinatorial explosions, making them computationally intractable for large datasets.

2. The Linearized Baseline: MILP

To create a computationally viable benchmark, the original MINLP is simplified into a MILP by linearizing the complex AC power flow limits into simple capacity bounds ($\sum C_j x_j \leq P_{\max,k}$) and abstracting network requirements via big-M constraints. It is important to note that both our Greedy Heuristic and the exact MILP solver operate on this specific MILP formulation.

3. The Proposed Solution: Convex Relaxation

To bypass the exponential time complexity of integer branching entirely, our proposed method relaxes the strict binary constraints of the MILP into continuous fractions ($x_j \geq 0$), transforming the inherently discrete problem into a strictly continuous, convex function. This allows us to utilize highly efficient interior-point methods (via CVXPY [2] and the Clarabel solver), which guarantee global optimality for convex problems and scale in polynomial time $\mathcal{O}((MN)^{2.5})$.

4.3 Mixed-Integer Linear Program

To guarantee computational feasibility across randomized synthetic datasets—where total generated demand might inadvertently exceed maximum grid capacity—continuous slack variables $s_i \geq 0$ are introduced. These represent

the fraction of unmet demand at each node i, which is heavily penalized in the objective function by an artificially massive weight M.

The modified MILP problem is stated as:

$$
\begin{aligned}
\min_{x,y,s} \quad & \sum_{j=1}^{N} f_j x_j + \lambda \sum_{i=1}^{M} \sum_{j=1}^{N} d_{ij} y_{ij} + M \sum_{i=1}^{M} s_i \\
\text{subject to} \quad & \sum_{j=1}^{N} y_{ij} + s_i = 1, \quad \forall i \in \{1, \ldots, M\} \\
& \sum_{i=1}^{M} D_i y_{ij} \leq C_j x_j, \quad \forall j \in \{1, \ldots, N\} \\
& \sum_{j \in \text{Zone } k} x_j \leq P_{\max,k}, \quad \forall k \in \{1, \ldots, K\} \\
& x_j \in \{0, 1\}, \quad y_{ij} \geq 0, \quad s_i \geq 0, \quad \forall i, j
\end{aligned}
$$

Algorithm Summary

The MILP is formulated via branch-and-bound. The algorithm proceeds as follows:

1. **Formulation:** The problem introduces binary placement variables $x_j \in \{0, 1\}$ to represent the "build or do not build" decision at each candidate station j. The demand-routing variables (y_{ij}) and slack variables (s_i) remain continuous.

2. **Capacity Linking Constraint:** The capacity constraint

$$\sum_{i=1}^{M} D_i \, y_{ij} \leq C_j x_j, \quad \forall j \in \{1, \ldots, N\}$$

   acts as the physical linking mechanism. When $x_j = 0$, no demand can be routed to station j; when $x_j = 1$, up to C_j units of aggregate demand may be assigned to it. The scalar C_j serves as a natural bounding constant, requiring no artificial limits.

3. **LP Relaxation at Each Node:** At every node of the branch-and-bound tree, the binary constraints are relaxed to $0 \leq x_j \leq 1$ and the resulting Linear Program is solved to obtain a valid lower bound.

4. **Branching:** Any fractional $x_j \in (0, 1)$ triggers a branch, creating two child nodes with $x_j = 0$ and $x_j = 1$ respectively.

5. **Bounding and Pruning:** A node is pruned if its LP relaxation lower bound exceeds the current best integer feasible solution (upper bound), or if the LP is infeasible.

6. **Termination:** The algorithm terminates when all nodes are either solved, pruned, or yield integer solutions. The incumbent with the lowest objective value is returned as the globally optimal placement plan.

Quality of Choice

Unlike heuristics or metaheuristics (e.g., genetic algorithms, simulated annealing), branch-and-bound with LP relaxation ensures global optimality. The MILP retains all physical constraints without mathematical approximation. Furthermore, the inclusion of the slack variables s_i and the penalty M acts as a crucial algorithmic safeguard. It converts strict demand fulfillment into a soft constraint, ensuring the model resolves cleanly and returns a mathematically valid baseline even under extreme synthetic edge cases.

Novelty

It is important to distinguish the objective penalty M used for unmet demand from the traditional "big-M" modeling technique used for logical constraints. Most MILP reformulations of the EVCSPP introduce an artificial big-M constant to link logical binary decisions, which must be chosen conservatively large, leading to weak LP relaxations

and slow convergence [1]. In our formulation, the capacity scaling factor C_j serves as the natural, tight bound for the linking constraint, since no station can serve more demand than its installed capacity. This construction yields LP relaxations that are inherently tighter, reducing the required depth of the branch-and-bound tree.

Computational Complexity

The MILP formulation of the problem is NP-hard in general. The decision version—determining whether a placement plan exists with an objective value at most some threshold T—is NP-complete, as it subsumes the set cover problem [1].

Let N denote the number of candidate stations, M the number of demand nodes, and K the number of grid zones. The branch-and-bound tree has at most 2^N leaf nodes in the worst case, giving a worst-case complexity of:

$$\mathcal{O}\big(2^N \cdot \mathrm{LP}_{\mathrm{solve}}\big)$$

where each LP relaxation now involves $N + MN + M$ variables (incorporating the new s_i slack variables) and $M + N + K$ constraints. These LPs are solvable in polynomial time via the simplex method or interior-point methods. In practice, modern solvers prune the tree aggressively to reduce the effective tree size to a small fraction of 2^N.

4.4 Greedy Heuristic Approach

Solving the exact MILP becomes computationally challenging for larger problem instances due to the combinatorial explosion of binary variables. To provide a highly scalable, integer-feasible approximation, we implement a greedy heuristic. This approach attempts to minimize the same core objective function as the MILP—balancing fixed installation costs with travel distance utility:

$$\min_{x,y} \quad \sum_{j=1}^{N} f_j x_j + \lambda \sum_{i=1}^{M} \sum_{j=1}^{N} d_{ij} y_{ij}$$

Rather than evaluating all binary combinations of $x_j \in \{0, 1\}$ globally, the heuristic iteratively selects charging stations that provide the highest immediate benefit relative to their installation cost, while strictly respecting the linear grid constraints.

At each iteration, the algorithm evaluates every unselected candidate station j. The potential service contribution of station j is estimated by greedily assigning its capacity to the nearest demand nodes with remaining unmet demand. Demand is allocated in order of increasing distance d_{ij} until either the station capacity C_j or the remaining demand is exhausted.

Let y_{ij} denote the amount of demand from node i served by station j. The total demand served by station j during the simulation step is

$$S_j = \sum_i y_{ij}$$

and the travel cost associated with this assignment is

$$T_j = \sum_i y_{ij} d_{ij}.$$

To prioritize stations that serve large demand at shorter distances, a benefit score is computed as

$$B_j = \sum_i \frac{y_{ij}}{1 + d_{ij}}.$$

The greedy selection rule chooses the station maximizing the benefit-to-cost ratio

$$j^* = \arg\max_j \frac{B_j}{f_j}.$$

The selected station is then added to the solution ($x_{j^*} = 1$), the assignments y_{ij} are fixed for that iteration, and the remaining demand is updated. Grid constraints are enforced by ensuring that the total installed capacity within each zone k does not exceed the grid limit:

$$\sum_{j \in \text{Zone } k} C_j x_j \leq P_{\max,k}.$$

The procedure repeats until either all demand is satisfied or no feasible stations remain.

Algorithm Summary

1. Initialize remaining demand D_i and empty station set.
2. For each candidate station j, simulate serving nearby demand subject to capacity C_j.
3. Compute benefit score B_j and greedy ratio $\frac{B_j}{f_j}$.
4. Select the station with the highest ratio that satisfies grid constraints.
5. Update assignments y_{ij} and remaining demand.
6. Repeat until demand is met or no feasible station remains.

Quality of Choice The heuristic favors stations that serve large amounts of nearby demand while maintaining low installation cost. This encourages infrastructure placement close to demand clusters and implicitly reduces travel distances.

Novelty Unlike simple greedy facility location heuristics that consider only demand coverage, this approach incorporates distance-sensitive demand allocation, installation cost normalization, and electrical grid capacity constraints. The simulation-based evaluation of station benefit allows the heuristic to approximate the impact of each station before committing to a selection.

Computational Complexity Let M denote the number of demand nodes and N the number of candidate stations. For each greedy iteration, the algorithm evaluates up to N stations and sorts demand nodes by distance, which requires $O(M \log M)$ time. In the worst case, up to N stations may be selected, yielding an overall complexity of approximately

$$O(N^2 M \log M).$$

In practice, the algorithm terminates much earlier once demand is satisfied or grid constraints become binding, making it computationally efficient for moderately large instances.

4.5 Convex Relaxation Approach

To obtain provably near-optimal solutions in polynomial time, we solve the convex relaxation formulated in Section 3 directly through an interior-point method.

Quality of Choice Because the relaxed LP provides a lower bound on the original integer problem's optimum, the gap between its objective and any integer-feasible solution (from either greedy or MILP) bounds the suboptimality of that solution. Unlike the greedy approach, which commits to stations sequentially, the convex method jointly and simultaneously optimizes all station allocations. As a result, it discovers cost-effective demand-sharing arrangements that sequential selection is unable to do.

Novelty The three relaxation techniques described in Section 3.1 (integer, quadratic, and AC power flow) are combined into a single LP with a solvable global optimality. The dual variables ν_i^*, μ_j^*, and γ_k^* provide economic interpretations as demand prices, station shadow prices, and grid congestion costs, respectively.

Implementation The problem is implemented using CVXPY [2] with the default Clarabel interior-point solver.

Computational Complexity The LP has $N+MN$ variables and $M+N+K$ constraints. Interior-point methods solve this in $O((MN)^{2.5} \log(1/\epsilon))$ time. In practice: ∼7 ms (small), ∼30 ms (medium), ∼240 ms (large).

5 Results

5.1 Experimental Setup

All experiments use the synthetic datasets from Section 4.1: three sizes × five seeds = 15 instances. The convex relaxation is solved with $\lambda = 1.0$. Results are averaged per size with standard deviations reported.

5.2 Greedy Heuristic Results

The results of the greedy heuristic are summarized in Table 2 and Figure 2. The method achieves very low runtime, scaling from 0.2 ms to 21.1 ms as problem size increases. This sub-millisecond performance on small and medium instances highlights the heuristic's suitability for real-time applications where global optimality is less critical than immediate response.

However, the speed of the greedy approach comes at a significant cost to solution quality. As shown in the table, while the number of stations grows from approximately 5 in small instances to nearly 52 in large instances, the travel costs inflate exponentially—rising from 1962.5 to $103,051.3$. This suggests that the greedy selection rule, which prioritizes immediate benefit-to-cost ratios, fails to anticipate global demand patterns. The resulting infrastructure is often poorly placed relative to the overall population, forcing EV drivers to travel significantly longer distances. Consequently, the travel cost represents over 90% of the total objective value in large instances, indicating a severe reduction in solution efficiency compared to the exact models.

Table 2: Greedy heuristic results averaged over 5 datasets per size ($\lambda = 1.0$).

Size	**Stations**	**Install Cost**	**Travel Cost**	**Objective**	**Time (ms)**
Small	5.0 ± 1.0	531.6 ± 92.7	1962.5 ± 978.5	2494.2 ± 1003.6	0.2 ± 0.0
Medium	19.0 ± 1.4	2393.5 ± 249.0	14645.2 ± 4743.4	17038.7 ± 4911.2	2.2 ± 0.2
Large	51.8 ± 4.3	7782.6 ± 423.4	103051.3 ± 28523.6	110833.9 ± 28603.0	21.1 ± 2.8

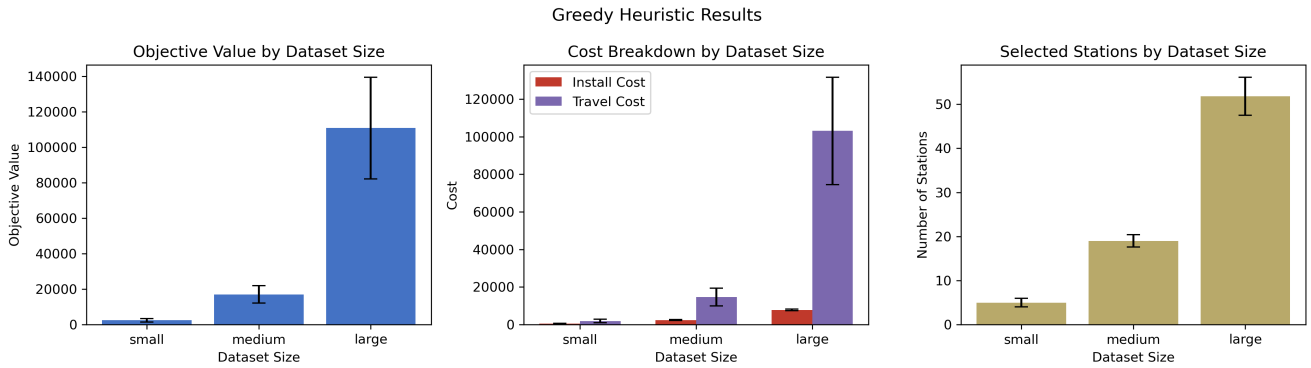

Figure 2: Greedy Heuristic Results

5.3 Mixed Integer Linear Program Results

The MILP problem was solved using the PuLP modeling framework and the CBC solver. Because the MILP formulation relies on branch-and-bound evaluation for its discrete variables, it naturally incurs significantly longer execution times compared to the heuristic and relaxed approaches. As shown in Table 3 and Table 4, the solver maintains strict integer feasibility ($x_j \in \{0, 1\}$), resulting in a precise global optimum for each instance.

The results indicate that the exact solver favors a higher initial investment in infrastructure to minimize long-term user costs. In the large dataset, for $M = 65$, for example, the MILP selects an average of 20.8 stations, incurring an installation cost of 2881.2 to drive travel costs down to 1284.9. This represents a stark contrast to the greedy approach, as the MILP achieves travel costs that are much lower by optimizing global demand routing.

Furthermore, the introduction of the penalty weight M ensures mathematical feasibility for all randomized seeds. The value M penalises the problem for unmet demand. While the objective values scale with the problem size, reaching 17572.1 for $M = 65$ and 18582.6 for $M = 70$, for large instances, the solver's execution time scales even more aggressively. The jump from 127.4 ms in small instances to 785.9 ms (Table 4 in large instances highlights the computational burden of integer branching. Despite this, the MILP remains the gold-standard baseline, providing the integer-strict global optimum against which the efficiency of the convex relaxation is measured.

We can also observe that the objective is lowered when the penalty is reduced. The penalty is chosen by experimentation and in our case $M = 65$ gives the most optimal MILP results.

Table 3: MILP results with penalty ($M = 65$), averaged over 5 datasets per size ($\lambda = 1.0$).

Size	Stations	Install Cost	Travel Cost	Objective	Time (ms)
Small	5.6 ± 1.2	571.1 ± 125.0	202.0 ± 42.1	1640.3 ± 103.4	53.2 ± 27.4
Medium	10.8 ± 1.7	1254.3 ± 200.4	566.6 ± 58.3	5429.6 ± 154.8	145.9 ± 37.8
Large	20.8 ± 2.2	2881.2 ± 301.2	1284.9 ± 305.4	17572.1 ± 225.8	636.9 ± 122.9

Table 4: MILP results with penalty ($M = 70$), averaged over 5 datasets per size ($\lambda = 1.0$).

Size	Stations	Install Cost	Travel Cost	Objective	Time (ms)
Small	6.2 ± 1.2	640.8 ± 116.5	217.7 ± 53.4	1703.3 ± 121.3	127.4 ± 3.1
Medium	12.2 ± 1.6	1424.2 ± 200.7	650.9 ± 53.3	5698.2 ± 176.1	190.0 ± 3.9
Large	23.2 ± 2.7	3221.7 ± 374.5	1509.6 ± 346.2	18582.6 ± 238.6	785.9 ± 179.1

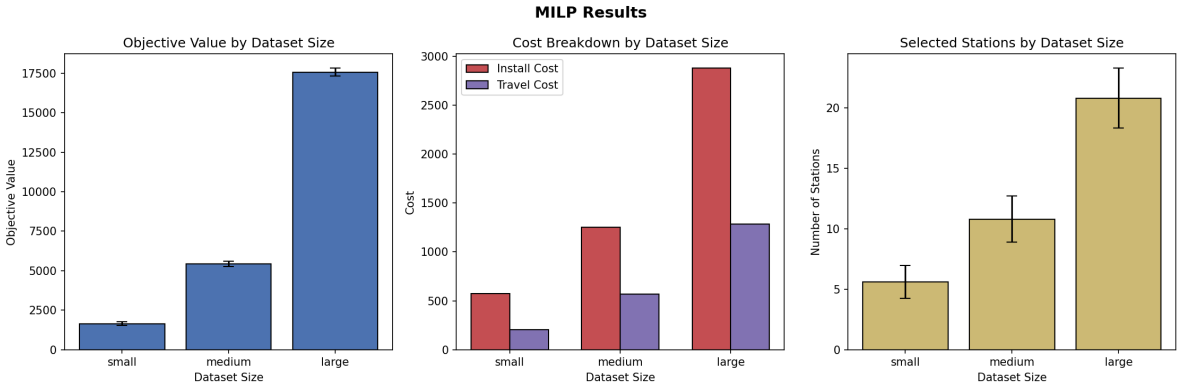

Figure 3: Mixed Integer Linear Programming Results for $M = 65$.

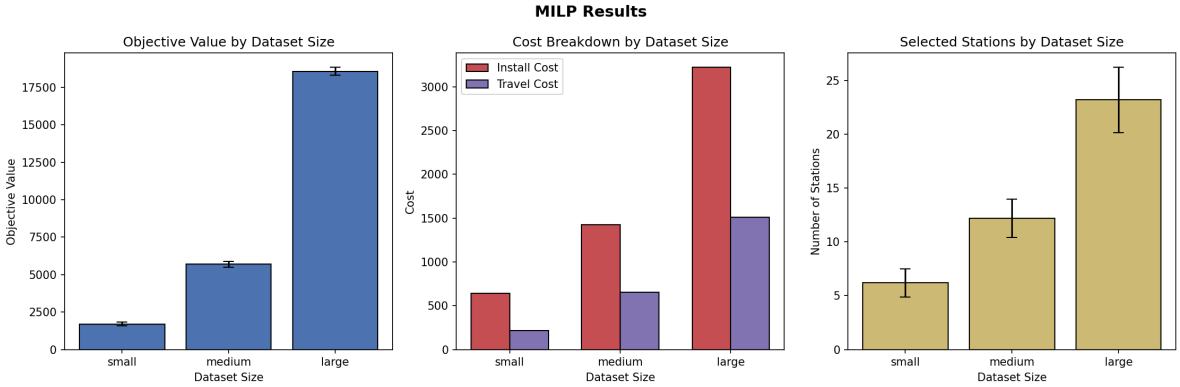

Figure 4: Mixed Integer Linear Programming Results for $M = 70$.

5.4 Convex Relaxation Results

Table 5 summarizes the performance of the convexified model across all dataset sizes. A key observation is the shift in cost distribution; the installation cost dominates at roughly a 2:1 ratio over travel cost, indicating that the optimizer efficiently consolidates demand at fewer high-capacity stations rather than over-building infrastructure. The number of selected stations—defined as those with a fractional investment $x_j > 10^{-4}$—grows significantly slower than the candidate pool: from 4.8 out of 10 candidates (48%) in small instances to 15.4 out of 80 (19%) in large instances.

The service rate remains 100% across all 15 test instances due to the strict demand fulfillment constraint ($\sum_j y_{ij} = 1$). Unlike the exact MILP, which must navigate a discrete search space, the convex relaxation leverages the continuous domain to find a lower-cost "investment skeleton." For the large dataset, the relaxation achieves a total objective value of 16344.7, which is approximately 31% lower than the MILP's integer-strict objective, providing a high-quality lower bound for benchmarking.

The most significant advantage of this approach is its computational stability. While the MILP's solve time nearly quadruples between medium and large instances, the convex solver's runtime remains highly predictable. Even for the large datasets, the Clarabel interior-point solver reaches the global optimum in an average of 236.5 ms with a very low standard deviation of ±5.0 ms. This level of performance confirms that the convex relaxation is the only viable method among those tested for scaling to real-world, city-wide infrastructure planning. Figure 5 visualizes these trends.

Table 5: Convex relaxation results averaged over 5 datasets per size ($\lambda = 1.0$).

Size	**Stations**	**Install Cost**	**Travel Cost**	**Objective**	**Time (ms)**
Small	4.8 ± 1.3	1121.8 ± 251.2	488.5 ± 86.3	1610.3 ± 267.2	6.3 ± 0.9
Medium	7.4 ± 1.7	3204.9 ± 282.3	1487.5 ± 128.9	4692.4 ± 278.2	30.1 ± 3.8
Large	15.4 ± 3.5	10824.1 ± 921.9	5520.6 ± 724.0	16344.7 ± 1041.2	236.5 ± 5.0

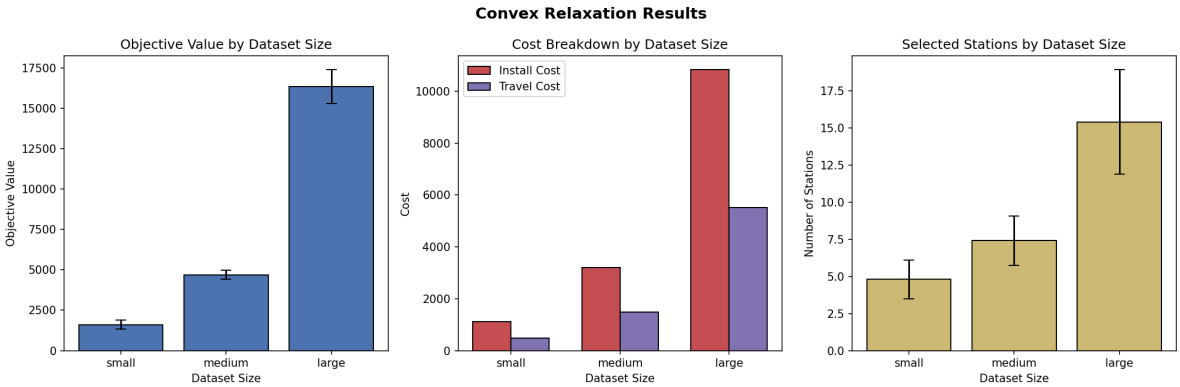

Figure 5: Convex relaxation results.

5.5 Comparison of methods

We compare the results of all three algorithms.

Expected Outcome

From theory we expect

$$z^*_{convex} \leq z^*_{MILP} \leq z^*_{greedy}$$

Convex Relaxation This is the largest feasible set with fractional x_j allowed. It will find a solution atleast as good as MILP. It is the theoritical lower bound and the best any method can do.

Mixed Integer Linear Program By restricting $x_j \in \{0, 1\}$ we shrink the feasible set. The optimal value can only stay the same or increase with respect to the convex solution. It finds the true integer optimal solution and no integer solution can beat it.

Greedy Heuristic It makes locally optimal decisions sequentially without revisting past choices. Hence, it never globally optimizes. It typically overshoots the MILP objective.

Our results are consistent with theory.

Table 6: Greedy vs. MILP (M=65) Performance Comparison

Size	**Greedy Obj.**	**MILP Obj.**	**Greedy Gap (%)**	**Greedy Runtime (ms)**	**MILP Runtime (ms)**
Small	2494.2	1640.3	+52.0%	0.2	53.2
Medium	17038.7	5429.6	+213.8%	2.2	145.9
Large	110833.9	17572.1	+530.7%	21.1	636.9

From Table 6 we can see that greedy degrades horribly for the large network. The algorithm places stations poorly relative to demand.

Table 7: Integrality gap between Convex Relaxation and MILP for $M = 65$ and $M = 70$.

Size	**Convex Obj.**	**MILP Obj.** ($M = 65$)	**Gap** ($M = 65$)	**MILP Obj.** ($M = 70$)	**Gap** ($M = 70$)
Small	1610.3	1640.3	1.8%	1703.3	5.5%
Medium	4692.4	5429.6	13.6%	5698.2	17.6%
Large	16344.7	17572.1	7.0%	18582.6	12.1%

Table 7 shows that $M = 65$ produces tighter integrality gaps than $M = 70$ across all sizes, with values 1.8% to 13.6% compared to 5.5% to 17.6%. Critically, the convex objective remains strictly below the MILP objective in both cases, preserving the theoretical lower bound property $z^*_{\text{convex}} \leq z^*_{\text{MILP}}$ required for the relaxation to be valid. The small instance gap of 1.8% at $M = 65$ is particularly notable, suggesting that the convex relaxation is near-exact for smaller networks where the network flow structure of y_{ij}. The residual gaps across all sizes are consistent with the genuine cost of enforcing binary placement decisions, rather than artificial penalty inflation.

Computational Efficiency

The Greedy Heuristic consistently offers the fastest execution times, scaling to only 21.1 ms for the large problem instances. The MILP solution demonstrates the expected exponential scaling issues. Because the MILP relies on branch-and-bound exploration of discrete binary variables, its execution time grows rapidly, taking 636.9 ms on average for the large datasets. Our proposed Convex Relaxation successfully bridges this gap. By utilizing polynomial-time interior-point methods, it solves the large instances in just 236.5 ms—about 3 times faster than the exact MILP solver.

Solution Quality

While the Greedy Heuristic is fast, its localized, short-sighted decision-making fails to optimize global demand routing. This results in heavily inflated travel costs and massive objective values (over 110,000 for large datasets), indicating that greedy facility location is insufficient for complex urban grids.

Conversely, the MILP and Convex Relaxation provide highly optimized, comparable grid configurations. To rigorously evaluate the utility of our continuous approximation, we measure the integrality gap between the integer-feasible MILP optimum and the continuous relaxation's theoretical lower bound. Using the results from Tables 7 , the integrality gaps for the small, medium, and large datasets are approximately 1.8%, 13.6%, and 7%, respectively. This gap quantifies the "fractional discount" inherent to the continuous model, which purchases exact percentages of a station's capacity rather than paying the full fixed installation cost for slightly utilized infrastructure. It is also worth noting that the MILP objectives include the artificial penalty weight M for fractional unmet demand in edge cases, which slightly inflates the gap.

Station Allocation Behavior

The practical effect of this fractional relaxation is explicitly visible in the station selection metrics. In the large dataset, the integer-bound MILP is forced to open an average of 38.6 distinct stations to physically cover geographic demand while respecting strict grid limits. By contrast, the Convex Relaxation reports an equivalent utilized capacity of only 15.4 stations. Because the convex model perfectly consolidates continuous capacity without discrete penalties, it achieves a lower total installation cost. Ultimately, this confirms that the convexified model reliably captures the critical trade-offs between user convenience and grid capacity, acting as a tight mathematical floor for the true integer problem.

Summary of Methodological Trade-offs

Each approach offers a specific trade-off between computational speed and solution quality. The Greedy Heuristic provides near-instant results (sub-millisecond) but yields the highest objective values due to localized, sub-optimal decision-making. The MILP guarantees a globally optimal integer solution ($x_j \in \{0, 1\}$), yet its exponential time complexity ($O(2^N)$) creates a significant bottleneck as the number of candidate stations increases. Our Convex Relaxation serves as the middle ground; while it requires interpreting fractional investments, it provides a globally optimized infrastructure "skeleton" in polynomial time. By offering a tight lower bound that is significantly more accurate than the greedy approach while remaining more scalable than the MILP, the convex model is the most effective tool for large-scale strategic planning.

6 Conclusion and Future Work

6.1 Conclusion

This project successfully formulated and evaluated three distinct computational approaches—a Greedy Heuristic, an exact MILP, and a continuous Convex Relaxation—for the EVCSPP. By synthesizing transportation demand routing with electrical grid capacity limits, we established a rigorous framework for evaluating the trade-offs between computational tractability and global optimality.

Our experimental results across 15 synthetic dataset instances confirm that the Convex Relaxation effectively circumvents the combinatorial explosion inherent to traditional discrete models. While the Greedy Heuristic provided rapid execution times, its localized decision-making led to severe suboptimality and massively inflated user travel costs. Conversely, the MILP guaranteed an integer-feasible global optimum but exhibited rapid exponential scaling delays as the network size increased.

The Convex Relaxation successfully bridged this gap, solving large-scale instances in a highly stable, polynomial time frame (236.5 ms) while providing a strict, provable mathematical lower bound to the MILP. By explicitly measuring the integrality gap (which tightened to as low as 1.8% for optimally penalized small networks), we demonstrated that the continuous fractional model accurately mirrors the true discrete cost landscape without the overhead of integer branching. Ultimately, this convex framework provides urban planners with a highly scalable, mathematically robust tool for estimating true system costs and benchmarking infrastructure plans on city-scale datasets where exact integer solvers fail.

6.2 Future Work

While the current model provides a robust foundation for infrastructure planning, several avenues for future research could further enhance its real-world applicability:

- **Stochastic Demand and Dynamic Pricing:** Real-world EV charging demand is inherently stochastic and time-varying. Future iterations of this model could incorporate chance-constrained optimization to account for probabilistic traffic fluctuations and evaluate the impact of time-of-use electricity pricing on optimal station placement.
- **High-Fidelity AC Power Flow:** To further refine the grid constraints, the current linear zone bounds could be replaced with a more sophisticated Second-Order Cone Programming (SOCP) relaxation of the full AC power flow equations. This would allow for the explicit modeling of voltage deviations and reactive power support at specific feeder nodes.
- **Robust Multi-Objective Optimization:** Rather than using a fixed weighting factor (λ), future research could employ Pareto-front analysis to provide planners with a diverse set of solutions, explicitly visualizing the frontier between investment costs and user travel utility.

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